

INDUWARA DINUSITH WIJESINGHE

Undergraduate Mechanical Engineering Student — Mechatronics Specialization

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LinkedIn | GitHub

Profile

Third-year Mechanical Engineering undergraduate at the University of Moratuwa (2023 intake), specializing in the Mechatronics stream, with hands-on experience in robotics, mechanical design, simulation, embedded systems, and automation. Experienced in multidisciplinary engineering projects using SolidWorks, AutoCAD, MATLAB/Simulink, Simscape Multibody, Python, C++, Raspberry Pi, Arduino/ESP32, and computer vision. Seeking an engineering internship to apply knowledge of mechanical engineering, mechatronics, robotics, control, and simulation to practical engineering challenges.

Education

BSc (Hons) in Mechanical Engineering (Mechatronics Specialization)

2024–Present

University of Moratuwa

GPA: 3.82 / 4.00

Relevant coursework: Mechatronics, Robotics, Automation, Mechanical Design, Engineering Mechanics, Dynamics and Control, Manufacturing.

Nalanda College, Colombo 10

Advanced Level (2022): Z-score 2.1310 — Combined Mathematics A, Physics A, Chemistry A, General English A.

Ordinary Level (2019): 9 A's (Distinction passes).

Technical Skills

Mechanical / Mechatronics: Mechanical design, CAD modelling, kinematics, kinetics, fluid machinery fundamentals, robotics, system modelling, engineering problem solving

Simulation / Engineering: SolidWorks, AutoCAD, MATLAB, Simscape Multibody, COMSOL Multiphysics, MSC Adams, Ansys

Programming / Computer Vision: Python, C++, MATLAB programming, OpenCV, YOLO, basic image processing and computer vision

Embedded / Hardware: Raspberry Pi, ESP32, Arduino, sensors

Transferable: Teamwork, research, presentation, time management, problem solving

Languages: English (fluent), Sinhala (native)

Engineering Projects

SAUVC 2026 — Autonomous Underwater Vehicle

IMU, Depth Control, ESP32/Firebase

- Contributed to development of an AUV for the SAUVC 2026 competition, focusing on underwater sensing, orientation, depth control, and communication.
- Integrated MPU9250/MPU6500 IMU and QMC5883L magnetometer for pitch, roll, yaw, and heading; used a pressure transmitter for depth control.
- Developed the ESP32/Firebase communication interface for transmitting control data to the onboard controller.

Vision-Guided Cartesian Pick-and-Place Robot

Raspberry Pi, YOLOv8, OpenCV, Arduino

- Designed and built a Cartesian pick-and-place robot with camera-based object detection and positioning.
- Developed and trained a YOLOv8 model to identify target objects and obtain their coordinates using Raspberry Pi and Python/OpenCV.
- Implemented NEMA 17 stepper motor control using Arduino and A4988 drivers for the turntable and robot axes.
- Built a web-based control interface using React, JavaScript, HTML, and CSS; designed critical mechanical parts in AutoCAD.

Smart Bicycle Instrumentation System

ESP32, IMU, GPS, Firebase IoT

- Designed an instrumentation system integrating heart-rate, IMU, GPS, and custom-built velocity sensors for real-time bicycle and rider data.
- Calibrated and filtered heart-rate sensor measurements using C++ to improve accuracy and reliability.
- Implemented GPS-based location tracking and ESP32 sensor data acquisition.
- Developed an IoT data pipeline and web GUI for real-time display of sensor data through Firebase.

Reverse Engineering of an Infrared Cooker

Structural & Material Analysis

- Conducted reverse engineering of a metallic cover assembly, including material identification, manufacturing-process evaluation, and design improvement.
- Identified low-carbon steel (AISI 1006–1010) using spark testing, density measurement, and Optical Emission Spectrometry.
- Modelled the cover, punch, and die in CAD and proposed reinforcement ribs and hemmed edges.

Multi-Purpose Device Holder

Manufacturing Technology

- Designed and fabricated a compact holder for a laptop, tablet, phone, and headset using wood and metal components.
- Applied woodworking and metalworking processes including cutting, drilling, mortise-and-tenon joinery, bending, and welding.
- Built an adjustable frame incorporating hinges and a wing-nut mechanism.

Biped Wall-Climbing Robot

Mechatronic System Design Project — Ongoing

- Designing a 5-DoF biped wall-climbing robot using suction-based adhesion for vertical wall inspection.
- Developing a negative-pressure adhesion system and two-legged inchworm gait for controlled vertical locomotion.
- Developing DH-based kinematic and inverse-kinematic models for joint control and suction-unit positioning.
- Integrating cameras, ToF sensors, encoders, IMU, and pressure sensors for perception, feedback, orientation, and adhesion monitoring.

6-DOF Robotic Arm Simulation

MATLAB/Simulink, Simscape Multibody — Ongoing

- Developing a simulation model of a 6-DOF robotic manipulator to study robot kinematics and multibody motion.

Professional Memberships & Achievements

- Student Affiliate Member — Institution of Mechanical Engineers (IMechE), 2025–Present (Membership No. 80794909).
- Dean’s List — University of Moratuwa, Faculty of Engineering (Semesters 2, 3, and 4).
- National Black Belt (1st Dan), Karate-do — Sri Lanka Karate-do Federation, awarded 2018.
- 2nd Place — Shotokan Karate-do National Championship Tournament, 2016.

Certifications

- Assured Diploma in Information Technology (DITEC) — ESOFT Metro Campus (Pearson Assured), 2020.
- MATLAB Onramp — MathWorks.
- Simulink Onramp — MathWorks.
- Diploma in Spoken English (Merit Pass) — British Way English Academy.

References

Referee Name

Title, Department, University / Company

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